



--	--	--	--	--	--	--	--	--

Student Number

2022 GLENWOOD HIGH SCHOOL

Year 11-Yearly Examination

Assessment Task 3

Mathematics Extension 1

**General
Instructions**

- * Reading Time – 10 minutes
- * Working time – 1.5 hours
- * Write using black pen
- * NESA approved calculators may be used
- * A reference sheet is provided
- * For questions in Section II, show relevant mathematical reasoning and/or calculations

**Total marks:
50**

Section I – 6 marks (pages 2 – 3)

- * Attempt Questions 1-6
- * Allow about 10 minutes for this section

Section II – 44 marks (pages 4 – 7)

- * Attempt Questions 7 – 9
- * Allow about 1 hours and 20 minutes for this section

Section I

6 marks

Attempt Questions 1 – 6.

Allow about 10 minutes for this section.

Use the multiple-choice answer sheet for Questions 1 – 6.

1. In how many distinct ways can the letters of the word POLLUTION be arranged?

- A. 4 578
- B. 6 890
- C. 90 720
- D. 362 880

2. What is the coefficient of x^4 in the expansion of $(2x - 1)^{10}$?

- A. -210
- B. 210
- C. 1 680
- D. 3 360

3. A function, $f(x)$, is said to be a self-inverse function if the inverse exists and

$$f(x) = f^{-1}(x).$$

Which of the following functions is considered a self-inverse function?

- A. $y = e^x$
- B. $y = x + 2$
- C. $y = x^2 + 2$
- D. $y = -x + 2$

4. The polynomial $P(x) = x^4 - kx^3 - 2x + 33$ has $(x - 3)$ as a factor. What is the value of k ?
- A. 4
- B. 5
- C. -4
- D. -5
5. A particle is moving in a straight line so that its displacement x after time t is given by $x = 2t^2 - \frac{1}{t}$. What is the acceleration of the particle after 1 second?
- A. 1 m/s²
- B. $1\frac{1}{2}$ m/s²
- C. 2 m/s²
- D. 5 m/s²
6. Mitchell wins a prize which consists of a bundle of fifty \$20 notes. He wants to share the money among himself and his six siblings.
- He does this by writing the seven names on cards which he places on a table. He then puts a \$20 note on each card, until he reaches the end of the row and then goes back to the start and repeats this process until all the notes are distributed.
- What is the most money that any one person will receive?
- A. \$140.00
- B. \$160.00
- C. \$180.00
- D. \$200.00

End of Section I

Section II

44 marks

Attempt Questions 7 – 9

Allow about 1 hour and 20 minutes for this section.

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 7 – 9, your responses should include relevant mathematical reasoning and/or calculations.

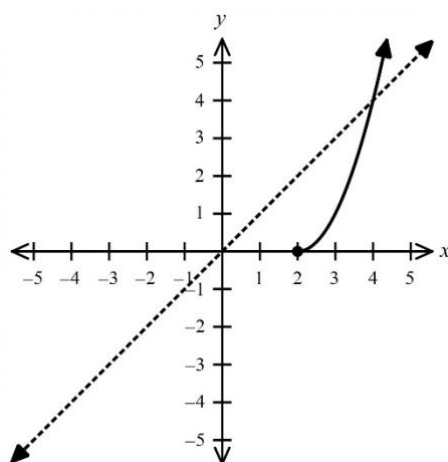
Question 7 (13 marks) Use a new writing booklet.

- a) Solve the inequality:

3

$$\frac{x}{5-x} \geq 1$$

- b) The graph of $y = f(x) = (x - 2)^2; [2, \infty)$ is shown below, along with the line $y = x$.



Write the equation of the inverse function in the form $y = f^{-1}(x)$ and state its domain.

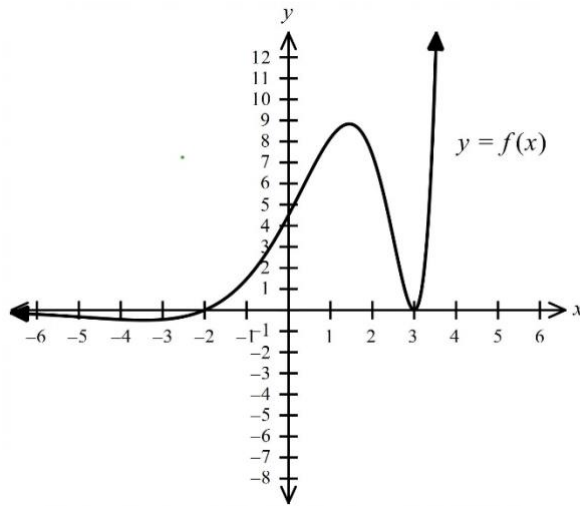
2

- c) Find the cartesian equation of the circle with the parametric equations

3

$$x = 10 + 4\cos\theta \text{ and } y = -8 + 4\sin\theta$$

d) The graph of $y = f(x)$ is shown below, for the domain $(-6, 6)$.



i) Draw a half-page sketch of $y = |(f(x))|$ showing important details.

1

ii) Draw a half-page sketch of $y = \frac{1}{f(x)}$ showing any asymptotes and intercepts.

2

e) Show that

$$\cos 20^\circ \sin 10^\circ = \frac{1 - 2 \sin 10^\circ}{4}$$

2

End of Question 7

Question 8 (17 marks) Use a new writing booklet.

- a) $P(x)$ has a remainder 2 when divided by $(x - 3)$ and a remainder -5 when divided by $(x - 2)$. 3

Find the remainder when $P(x)$ is divided by $x^2 - 5x + 6$.

- b) The displacement (in metres) from the origin, after t seconds, of a particle moving along the number line is given by:

$$x = t^3 - 10t^2 + 12t + 24.$$

- i) Find the times when the particle is at rest. 2
- ii) Find the velocity of the particle at the time where there is no accelerating force acting on it. 2
- c) i) State the domain and range of the function $f(x) = 3 \cos^{-1}(1 - 2x)$. 2
- ii) Hence, sketch $f(x) = 3 \cos^{-1}(1 - 2x)$. Label any intercepts with the coordinate axes. 2
- d) Eight differently coloured points are to be equally spaced around two identical circles. Each circle is to have four different coloured points. 2

How many different ways can these coloured points be arranged?

- e) Prove that 4

$$\frac{1 - \cos 2x}{1 + \cos 2x} = \tan^2 x$$

Hence, show that

$$\tan \frac{\pi}{12} = 2 - \sqrt{3}$$

End of Question 8

Question 9 (14 marks) Use a new writing booklet.

a) Show that $\sec(\tan^{-1} p) = \sqrt{1 + p^2}$. 2

b) If α, β, γ are the roots of the equation $2x^3 - 6x^2 + 10x - 5 = 0$, find the values of:

i) $\alpha + \beta + \gamma$ 1

ii) $\alpha\beta + \alpha\gamma + \beta\gamma$ 1

iii) $\alpha^2 + \beta^2 + \gamma^2$ 2

c) Show that for all integers $n \geq 3$, 3

$$\binom{n+2}{3} - \binom{n}{3} = n^2$$

d) The rate of change of the temperature T °C of a body after t minutes in an environment where the surrounding temperature is T_s °C is given by:

$$\frac{dT}{dt} = -k(T - T_s) \text{ where } k \text{ is some constant}$$

Garry takes out a small tub of ice-cream from the deep freezer which was set to -40 °C and finds that after one minute, the ice-cream warms to -30 °C.

Garry knows that the ideal temperature to enjoy his ice-cream is between -15 °C and -5 °C. The temperature in the room is kept at a constant 20 °C.

i) Show that the solution $T = T_s + Ae^{-kt}$, where A is some constant, satisfies the differential equation $\frac{dT}{dt} = -k(T - T_s)$. 1

ii) Find the exact values of A and k . 2

iii) Garry begins to eat his ice-cream when it is exactly -15 °C cold. How long does Garry have to enjoy his ice-cream before it becomes too warm for his liking. Give your answer correct to the nearest second. 2

End of Question 9

END OF PAPER

11X1 - Yearly 2022

1). 9 letters, 2 D's, 2 L's

$$\frac{9!}{2! \times 2!} = 90720 \quad (C)$$

2). $(2x-1)^{10}$

$$\begin{aligned} T_{k+1} &= \binom{10}{k} (2x)^{10-k} (-1)^k \\ &= \binom{10}{k} (2)^{10-k} (-1)^k x^{10-k} \end{aligned}$$

$$10-k = 4$$

$$k = 10-4 = 6$$

$$\begin{aligned} \text{Coefficient} &= \binom{10}{6} (2)^{10-6} (-1)^6 \\ &= \binom{10}{6} (2)^4 = 3360 \end{aligned} \quad (D)$$

3). $y = -x + 2$

Interchanging x & y , $x = -y + 2$

$$y = 2 - x$$

$\therefore (D)$

$$\begin{aligned} 4) \quad P(3) &= (3)^4 - k(3)^3 - 2(3) + 33 \\ &= 81 - 27k - 6 + 33 \\ &= 108 - 27k \end{aligned}$$

$$108 - 27k = 0$$

$$-27k = -108$$

$$k = \frac{-108}{-27} = 4 \quad (A)$$

$$5). \quad x = 2t^2 - \frac{1}{t}$$

$$v = 4t + \frac{1}{t^2}$$

$$a = 4 - \frac{2}{t^3}$$

$$\text{At } t=1, \quad a = 4 - \frac{2}{1} = 2 \quad \textcircled{C}$$

$$\textcircled{a} \text{ rest } v=0$$

$$\frac{1-2t^2}{1-t^2} = 0$$

$$1-2t^2 = 0$$

$$-2t^2 = -1$$

$$t^2 = \frac{1}{2}$$

$$t = \pm \frac{1}{\sqrt{2}} \quad \text{since } t \geq 0, \quad t = \frac{1}{\sqrt{2}} \quad \textcircled{C}$$

$$6). \quad \frac{50}{7} = 7.2$$

$\therefore$ Atleast one person should get 8 notes

$$8 \times 20 = 160 \quad \textcircled{B}$$

07)

$$a) \quad \frac{x}{5-x} \geq 1 \quad x \neq 5$$

$$\frac{x(5-x)^2}{5-x} \geq 1(5-x)^2$$

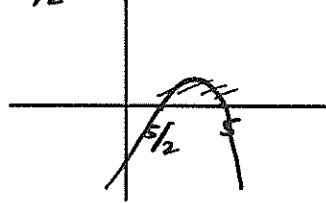
$$x(5-x) \geq (5-x)^2$$

$$x(5-x) - (5-x)^2 \geq 0$$

$$(5-x)(x - (5-x)) \geq 0$$

$$(5-x)(2x-5) \geq 0$$

$$\frac{5}{2} \leq x < 5$$



b). $f(x) = (x-2)^2; [2, \infty]$

Interchanging x & y

$$x = (y-2)^2$$

$$y-2 = \pm \sqrt{x}$$

$$y = \pm \sqrt{x} + 2$$

In the restricted domain

$$f^{-1}(x) = \sqrt{x} + 2$$

$$x \in [0, \infty)$$

c) $x = 10 + 4\cos z$

$$y = -8 + 4\sin z$$

Rearranging

$$\cos z = \frac{x-10}{4} \quad \text{--- (1)}$$

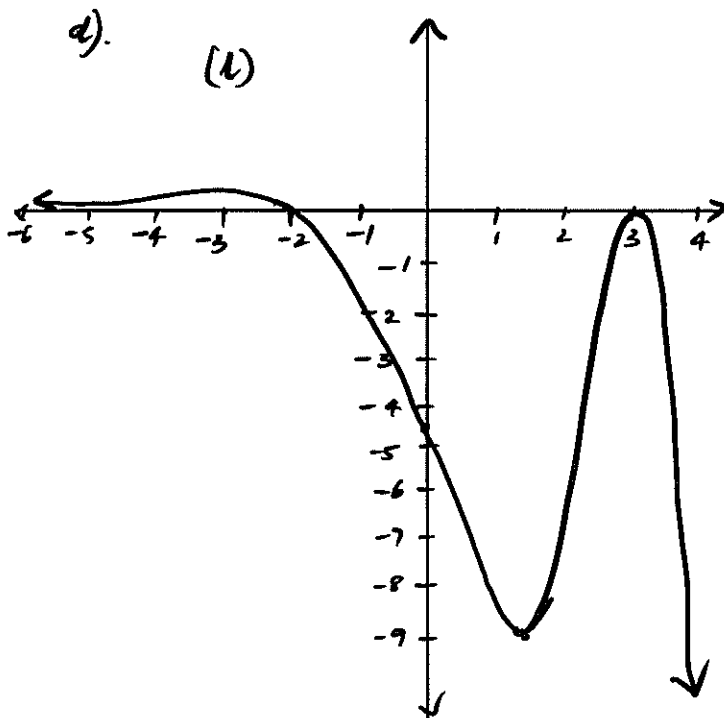
$$\sin z = \frac{y+8}{4} \quad \text{--- (2)}$$

$$\textcircled{1}^2 + \textcircled{2}^2$$

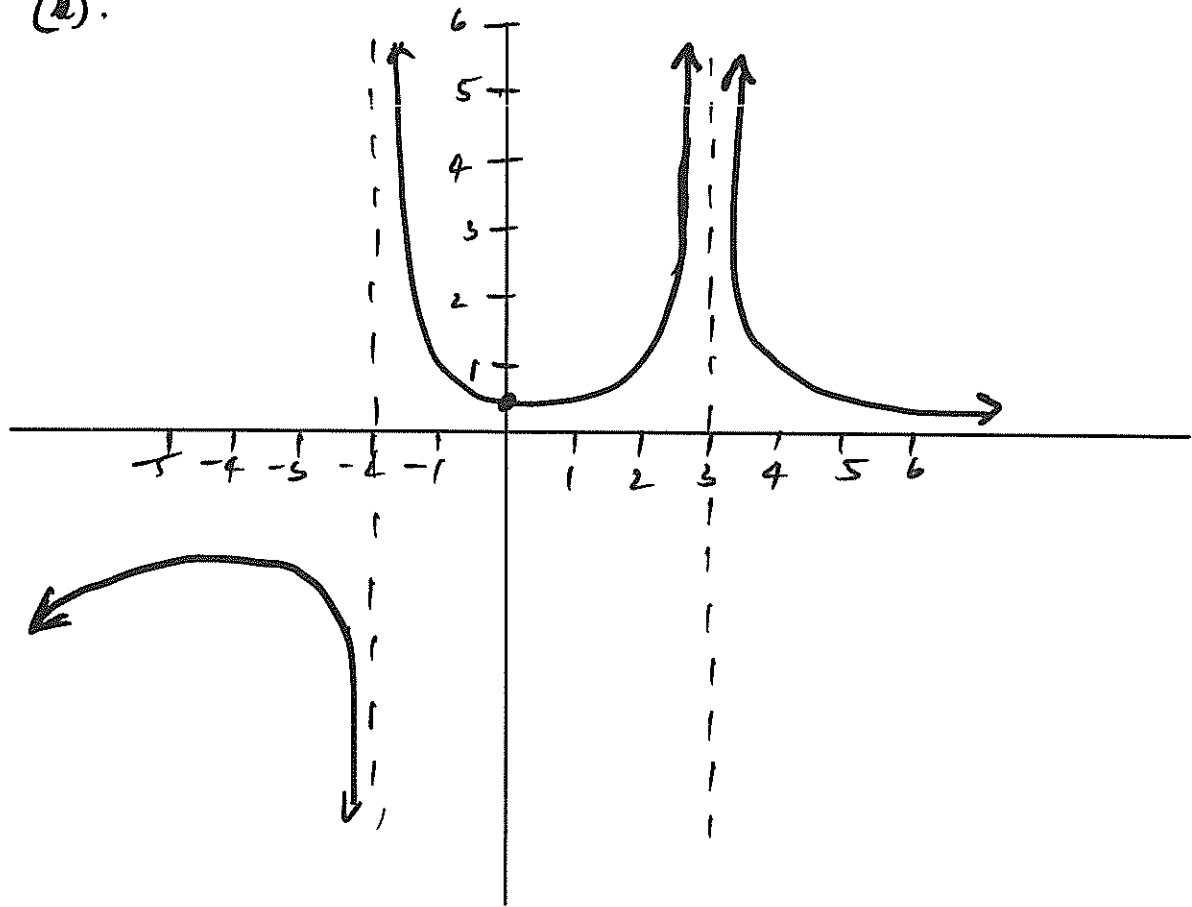
$$\sin^2 x + \cos^2 x = \left(\frac{y+8}{4}\right)^2 + \left(\frac{x-10}{4}\right)^2$$

$$1 = \frac{(y+8)^2 + (x-10)^2}{16}$$

$$\therefore (x-10)^2 + (y+8)^2 = 16$$



(11).



$$\begin{aligned} \text{c) } \cos 20 \sin 10 &= \frac{1}{2} (\sin 30 - \sin 10) \\ &= \frac{1}{2} \left(\frac{1}{2} - \sin 10 \right) \\ &= \frac{1}{2} \left(\frac{1 - 2 \sin 10}{2} \right) \\ &= \frac{1 - 2 \sin 10}{4} \end{aligned}$$

$$8). \quad P(3) = 2$$

$$P(2) = -5$$

$$P(x) = (x^2 - 5x + 6) Q(x) + ax + b$$

$$P(x) = (x-3)(x-2) Q(x) + ax + b$$

$$P(3) = a(3) + b = 2$$

$$3a + b = 2 \quad \text{--- (1)}$$

$$P(2) = a(2) + b = -5$$

$$2a + b = -5 \quad \text{--- (2)}$$

$$\textcircled{1} - \textcircled{2} \quad a = 7$$

$$b = 2 - 3(7)$$

$$= 2 - 21 = -19$$

$$\text{Remainder} = 7x - 19$$

$$b). \quad x = t^3 - 10t^2 + 12t + 24$$

$$v = 3t^2 - 20t + 12$$

$$\textcircled{a} \text{ rest, } v = 0$$

$$3t^2 - 20t + 12 = 0$$

$$(3t-2)(t-6) = 0$$

$$t = 2/3, 6$$

$$(u). \quad a = 6t - 20$$

$$\text{when } a = 0$$

$$\begin{matrix} 3t & \times & -2 \\ t & \times & -6 \end{matrix}$$

$$6t - 20 = 0$$

$$6t = 20$$

$$t = 20/6 = 10/3$$

$$\text{when } t = 10/3$$

$$v = 3\left(\frac{10}{3}\right)^2 - 20\left(\frac{10}{3}\right) + 12$$

$$= \frac{3(100)}{3} - \frac{200}{3} + 12$$

$$= -21\frac{1}{3}$$

c). (i) $y \in [0, 3\pi]$

$$-1 \leq 1 - 2x \leq 1$$

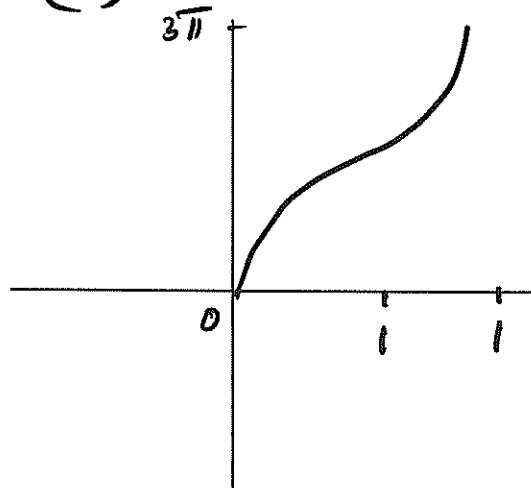
$$-2 \leq -2x \leq 0$$

$$\frac{-2}{-2} \geq x \geq \frac{0}{-2}$$

$$1 \geq x \geq 0$$

$$\therefore x \in [0, 1]$$

(ii)



d). $\frac{8C_4 \times 3! \times 3!}{2}$

Note:- $8C_4 \rightarrow$ choose 4 points from 8

$3! \rightarrow$ Arrange 4 points around circle

$3! \rightarrow$ Arrange the rest 4 around the circle

Divide by 2 as order of circles don't matter

$$c). \quad \frac{1 - \cos 2x}{1 + \cos 2x} = \tan^2 x$$

LHS

$$\frac{1 - (1 - 2\sin^2 x)}{1 + (2\cos^2 x - 1)}$$

$$= \frac{1 - 1 + 2\sin^2 x}{1 + 2\cos^2 x - 1}$$

$$= \frac{\cancel{2}\sin^2 x}{\cancel{2}\cos^2 x}$$

$$= \tan^2 x \quad \text{Q.E.D.}$$

$$\tan \frac{\pi}{12}$$

$$\text{Using } \tan^2 x = \frac{1 - \cos 2x}{1 + \cos 2x}$$

$$\left(\tan \frac{\pi}{12} \right)^2 = \frac{1 - \cos 2 \times \pi/12}{1 + \cos 2 \times \pi/12}$$

$$= \frac{1 - \cos \pi/6}{1 + \cos \pi/6}$$

$$= \frac{1 - \frac{\sqrt{3}}{2}}{1 + \frac{\sqrt{3}}{2}}$$

$$= \frac{\frac{2 - \sqrt{3}}{2}}{\frac{2 + \sqrt{3}}{2}}$$

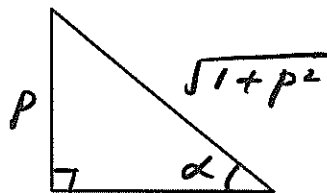
$$= \frac{(2 - \sqrt{3})(2 - \sqrt{3})}{(2 + \sqrt{3})(2 - \sqrt{3})}$$

$$= \frac{(2 - \sqrt{3})^2}{4 - 3} = (2 - \sqrt{3})^2$$

$$\tan \pi/12 = 2 - \sqrt{3} \quad \text{since } \tan \pi/12 > 0$$

89) a)

let $\alpha = \tan^{-1} p$



$$\sec(\tan^{-1} p) = \sec \alpha$$

$$= \frac{1}{\cos \alpha} = \frac{1}{\frac{1}{\sqrt{1+p^2}}} = \sqrt{1+p^2}$$

$$b) \quad \alpha + \beta + \gamma = \frac{-b}{a} = \frac{6}{2} = 3$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a} = \frac{10}{2} = 5$$

$$\begin{aligned} \alpha^2 + \beta^2 + \gamma^2 &= (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma) \\ &= (3)^2 - 2(5) \\ &= 9 - 10 \\ &= -1 \end{aligned}$$

$$c) \quad \binom{n+2}{3} - \binom{n}{3} = n^2$$

LHS

$$\binom{n+2}{3} - \binom{n}{3}$$

$$\frac{(n+2)!}{(n+2-3)! \times 3!} - \frac{n!}{(n-3)! \times 3!}$$

$$\frac{(n+2)!}{(n-1)! \times 3!} - \frac{n!}{(n-3)! \times 3!}$$

$$\frac{(n+2)(n+1)\cancel{n}(\cancel{n-1})!}{(\cancel{n-1})! \times 3!} - \frac{n(n-1)(n-2)(\cancel{n-3})!}{(\cancel{n-3})! \times 3!}$$

$$\begin{aligned}
& \frac{(n+2)(n+1)n - n(n-1)(n-2)}{3!} \\
&= \frac{n[(n^2+3n+2) - (n^2-3n+2)]}{3!} \\
&= \frac{n(\cancel{n^2} + 3n + \cancel{2} - \cancel{n^2} + 3n - \cancel{2})}{3!} \\
&= \frac{6n^2}{3!} \\
&= \frac{6n^2}{6} = n^2
\end{aligned}$$

$$d). \quad \frac{dT}{dt} = -k(T - T_s)$$

$$\text{At } t=0, \quad T = -40$$

$$t=1, \quad T = -30$$

$$T_s = 20^\circ$$

$$(1) \quad T = T_s + Ae^{-kt} \quad \text{--- (1)}$$

$$\frac{dT}{dt} = A e^{-kt} \times -k \quad \text{--- (2)}$$

$$\text{from (1)} \quad Ae^{-kt} = T - T_s$$

Sub in ②

$$\begin{aligned}\frac{dT}{dt} &= (T - T_s)x - k \\ &= -k(T - T_s)\end{aligned}$$

$$\begin{aligned}\text{(u)} \quad T &= T_s + Ae^{-kt} \\ T &= 20 + Ae^{-kt}\end{aligned}$$

$$\text{sub } t=0, \quad T = -40$$

$$-40 = 20 + Ae^0$$

$$-40 - 20 = A$$

$$A = -60$$

$$\therefore T = 20 - 60e^{-kt}$$

$$\text{sub } t=1, \quad T = -30$$

$$-30 = 20 - 60(e^{-k})$$

$$-50 = -60e^{-k}$$

$$\frac{-50}{-60} = e^{-k}$$

$$\ln \frac{5}{6} = \ln e^{-k}$$

$$\ln \frac{5}{6} = -k \ln e$$

$$k = -\ln 5/6 = \ln 6/5$$

$$\therefore T = 20 - 60 e^{(-\ln 6/5)t}$$

$$\text{At } T = -15$$

$$20 - 60 e^{(-\ln 6/5)t} = -15$$

$$- 60 e^{(-\ln 6/5)t} = -35$$

$$e^{(-\ln 6/5)t} = \frac{-35}{-60} = \frac{7}{12}$$

$$(-\ln 6/5)t = \ln 7/12$$

$$t = \frac{-1}{\ln 6/5} \times \ln 7/12$$

$$\text{At } T = -5$$

$$20 - 60 e^{(-\ln 6/5)t} = -5$$

$$- 60 e^{(-\ln 6/5)t} = -25$$

$$e^{(-\ln 4/5)t} = \frac{-25}{-60} = \frac{5}{12}$$

$$(-\ln 4/5)t = \ln 5/12$$

$$t = \frac{-1}{\ln 4/5} \times \ln 5/12$$

$$\text{Time Garry have} = \left(\frac{-1}{\ln 4/5} \times \ln 5/12 \right) - \left(\frac{-1}{\ln 4/5} \times \ln 7/12 \right)$$

$$= \frac{1}{\ln 4/5} \left(-\ln 5/12 + \ln 7/12 \right)$$

$$= 1 \text{ min } 51 \text{ sec}$$